GGG-M-125E/GEN
March 28, 1975
SUPERSEDING
GGG-M-125D
October 11, 1965, and
MIL-R-51057A(MU)
December 11, 1964

FEDERAL SPECIFICATION

RESPIRATOR AND RESPIRATOR ASSEMBLIES; AIR LINE; AIR FILTERING: POWERED:

(GENERAL SPECIFICATION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the general requirements for mask, hood, helmet respiratory protective systems, designed to protect wearers from specific requirements for the particular type, style, and models of masks and respiratory systems covered by the applicable detail specification ee 6.4). These respiratory protective systems are not intended to be used against toxic warfare agents, or in areas immediately dangerous to life and health.

1.2 Classification. Types, styles, and models of respirators shall be designated in the detailed specification for each respirator protective system.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

GGG-H-142 - Helmet, Construction Workers.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, and Standards, at the price indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Specifications:

MIL-E-5272 - Environmental Testing Aeronautical & Associated Equipment Specification For.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-202 - Test Method for Electronic & Electrical Component Parts.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

30CFR11 - Respiratory Protective Devices, Tests for Permissibilities; Fees.

(The Code of Federal Regulations (CFR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated reprints of certain regulations may be obtained from the Federal Agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

National Bureau of Standards (NBS)

C533 - Methods of Determining the Resolving Power of Photographic Lenses.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Approval and certification. The various types, styles, models, and air cleaners of respirator systems with their respective cartridges, filters, or air purifying units, specified in the detail specifications, shall be approved by the Mining Enforcement and Safety Administration (MESA) and the National Institute of Occupational Safety and Health (NIOSH), Department of Health Education and Welfare (HEW), as provided under the applicable parts of the CFR, Title 30, Part 11 (see 6.6). The M/NIOSH approval number shall appear on each respiratory system, or the approved part number shall appear.

3.1.1 Certification. The manufacturer shall submit evidence that the respirator has been tested and approved as required in 3.1.

3.2 Material. The material used in the manufacture and assembly of the respiratory equipment shall be new and of a quality to meet the requirements described herein and in the detail specifications.

3.3 Sanitization. When subjected to the sanitization test specified in 4.4.1, masks and helmets and accessory items, except disposable or single use types, shall show no visible signs of deterioration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

3.4 Lenses. Lenses shall be glass or plastic with smooth finished edges and corners. Lenses shall be free from inclusions, striae, waves, or other visible defects which may impair their optical quality. The prismatic effect caused by nonparallelism of the surfaces of lens shall not exceed 1/8-prism diopter. Lenses shall not be negative in refractive power in any meridian and the positive refractive power in any meridian shall not exceed 1/8 diopter. Lenses shall show clear definition when tested in accordance with 4.4.2.4, and shall transmit not less than 89 percent of the incident visible light. After subjection to the impact test of 4.4.2.6, lenses of solid plate glass or plastic shall not fracture or chip; lenses of laminated glass shall show no separation of glass from plastic interlayer.

3.5 Design. The design for the various respirator types, styles, and models, and air cleaners shall be as specified in the detailed specification (see 6.4).

3.6 Interchangeability. All respirator systems of the same classification furnished with similar options under a specific contract of the same manufacturer shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.7 Spare parts. When specified in the detail specification, spare parts shall be furnished for masks and helmet respirators (see 6.2.1).

3.8 Instructions. Each respirator system furnished under this specification shall be provided with instructions covering the use, care and operation of the respirator system, and a list of the component parts.

3.9 Marking. Facepieces, helmets, vests, cartridges, and filters of respiratory systems shall bear a permanent distinctive marking by which the manufacturer may be readily identified and shall show approval of the MESA and NIOSH. Elastomeric facepieces (reusable types) shall be marked with the quarter and year of manufacture.

3.10 Breathing tube leakage. When specified in detailed specification, the breathing tube shall not leak when subjected to the water immersion test in 4.4.3.

3.11 Helmet impact protection. When specified in detailed specification, the helmet shall meet the acceptance criteria specified in 4.4.4.

3.12 Centrifugal air cleaner requirements. When specified in detailed specification, the centrifugal air cleaner shall meet the acceptance criteria of the tests specified in 4.4.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

3.13 Workmanship. The quality of the workmanship shall be such as to produce respiratory systems that are in accordance with the requirements of this specification and the detail specification, and insure proper functioning of all parts of respirator assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material. Unless otherwise specified in the detailed specification the supplier shall certify that all materials used in the various types, styles, and models of respirators and respirator assemblies are new and unused.

4.2 Inspection lot. All respirator assemblies of the same type, style, and model offered to the Government at one time shall be considered a lot for purposes of inspection.

4.2.1 Qualify conformance inspection. Quality conformance inspection shall include the examination of 4.3 and the tests of 4.4 and the detail specification.

4.2.2 Examination. A random sample of respirators shall be selected from each lot in accordance with MIL-STD-105 at inspection level II. In terms of defects per 100 units, the Acceptable Quality Level shall be 2.5 for major defects and 6.5 minor defects.

4.2.3 Sampling for tests.

4.2.3.1 Sanitization test. When applicable, one respirator shall be selected at random from the lot. Failure to pass the assembly sanitization test specified in 4.4.1 shall be cause for rejection of the entire lot.

4.2.3.2 Lens test. When applicable, random sample consisting either of masks or respirators shall be selected in accordance with MIL-STD-105 at inspection level S-2. Failure of any lens to pass the tests of 4.4.2 shall be cause for rejection of the entire lot.

4.2.3.3 Leakage test. All breathing tubes shall be tested for leakage in accordance with 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

4.2.3.4 Helmet impact test. When applicable, one helmet respirator shall be selected at random from the lot and tested as specified in 4.4.4.

4.3 Examination. Respirator assemblies shall be examined for visual defects in accordance with detail specifications.

4.4 Tests. When specified in the detail specification, tests specified herein shall be performed.

4.4.1 Sanitization test. Sampled respirators, (disposable types excluded), selected in accordance with 4.2.3.1 shall be subjected to one of the following sanitization methods:

(a) Immersion for 10 minutes in a solution of formalin, made by placing one part of 40 percent formaldehyde solution into nine parts of water at room temperature.
(b) Subjection to a moist atmosphere of antiseptic gas, preferably formaldehyde, for a period of 10 minutes at room temperature.
(c) Subjection to use of a suitable sanitizing agent, such as a quaternary ammonium salt solution.

The sanitization test shall be repeated five times. Half-mask and full-mask respirators shall be rinsed and dried between each sanitizing treatment. Any signs of deterioration shall be cause for rejection.

4.4.1.1 Helmet sanitizing test. Sample helmets selected in accordance with 4.2.3.1, shall be subjected to the following test:

(a) All exterior and interior surfaces of helmet and other protective components, such as accessible vests, leather, or aluminized shrouds, shall be sprayed with a sanitizing detergent or wiped with a sponge dampened in a disinfecting detergent. After 10 minutes, the treated areas shall be wiped with a clean, damp sponge or cloth and allowed to dry. The above procedure shall be repeated five times. After the test, the treated areas shall show no signs of deterioration.
(b) The inner shrouds of helmets shall be removed and laundered 10 times by conventional laundering techniques. Shrouds shall show no signs of deterioration or difficulty in reinstalling in the helmet.

4.4.2 Lens tests. Lenses taken from sample masks or respirators, selected in accordance with 4.2.3.2, shall be tested for optical quality, prismatic power, refractive power, definition, visible transmissions, and impact resistance by methods specified in 4.4.2.1 through 4.4.2.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

4.4.2.1 Optical quality. The test for inclusions, striae, waves, and other defects which impair optical quality of the lens shall be conducted by examining the light pattern of a point source on a screen with lens interposed between the light source and the screen. (A shadowgraph or a similar instrument may be used in this test.)

4.4.2.2 Prismatic power. The lenses shall be tested for prismatic power with an eight-power telescope which has an effective aperture of 0.75 inch and is equipped with crosshairs in the focal plane of the ocular. The telescope shall be focused on an illuminated target, placed at a distance of 35 feet from the telescope objective, comprising a central dot and a concentric circle 1.0 inch in diameter. The telescope shall be so alined that the image of the central dot falls on the intersection of the crosshairs in the focal plane of the ocular. The lens shall be held in front of the objective lens of the telescope, and if the intersection point of the crosshairs falls outside the image of the circle, the prismatic power exceeds 1/8-prism diopter.

4.4.2.3 Refractive power. The lenses shall be tested for refractive power by any suitable instrument, such as a lensometer, vertometer, or telescope. The telescope method of measuring refractive power requires an external-focusing telescope with an effective aperture not less than 0.75 inch and with magnifying power not less than 8. The telescope shall be focused on an illuminated test chart (described in NBS circular C533) placed a distance of 35 feet from the telescope objective. The telescope shall be calibrated by successively locating the position of best focus with no lens in front of the objective, and with a standard lens of power +1/8 diopter. These positions shall be marked by scratches on the draw tube or other suitable index marks. The lens to be tested shall be held in front of the calibrated telescope, and if the position of the best focus falls outside the index marks, the refractive power is negative or in excess of +1/8 diopter.

4.4.2.4 Definition. The lenses shall be tested for definition with an eight-power telescope which has an effective aperture of 0.75 inch, and shall be focused at a distance of 35 feet on an illuminated test chart. As a test chart, the resolving power chart pattern 20 (see 6.3) of NBS circular C533 may be used. An advantage in adopting this chart is that, by its use, it becomes possible to provide all inspectors with identical chart, whereas if charts are improvised at different places, they are likely to be different. The pattern marked 20 should be clearly resolved with the target placed at a distance of 35 feet from the telescope objective used for testing the lenses.

4.4.2.5 Visible transmission. The standard source of radiant energy used in the measurement of the transmission of visible radiation (light) of lenses shall be a 500-watt (or other high-powered) gas-filled tungsten filament electric incandescent lamp, operated at rated voltage. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

visible radiation shall be determined photometrically by an observer, having normal color vision, as determined by the Holm-gren test for color vision, or with a physical photometer consisting of a thermopile (or other radiometer), and a luminosity solution having a spectral transmission curve which coincides with the visibility curve of the average eye.

4.4.2.6 Impact test. Lens shall be placed on a wooden block, recessed under the lens, except for a supporting margin 1/4-inch wide, with a 1/16-inch rubber gasket placed between the lens and supporting margin. A 7/8-inch diameter steel ball weighing approximately 1.56 ounces shall be dropped on the center of the lens from a height of 30 inches.

4.4.3 Leakage test. Each length of breathing tube shall be subjected to an internal air pressure of 5 pounds per square inch gage while the tube is immersed in water. No leakage shall occur while pressure is applied for a minimum of 10 seconds.

4.4.4 Helmet impact protection test. Helmets shall be tested for levels of both impact and penetration protection in accordance with the requirements of GGG-H-142.

4.4.5 Motor blower tests.

4.4.5.1 Explosion test. Explosion tests shall be performed in accordance with Procedure III, MIL-E-5272. Ignition of explosive mixtures shall not occur as a result of unit operation. Visual damage shall not be evident as a result of exposure.

4.4.5.2 Moisture resistance test. Moisture resistance tests shall be performed in accordance with method 106c, MIL-STD-202. No obvious degradation of air cleaner motor/blower subassembly shall be observed.

4.4.5.3 Sand and dust test. Sand and dust tests shall be performed in accordance with Procedure I, MIL-E-5272. Motor/blower subassembly shall operate properly after the test exposure and shall evidence no damage.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking of the respirators and components shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Masks and respirators. Respirators and components shall be packaged in a fiberboard box conforming to PPP-B-636, class weather-resistant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. Respirators and components of respirators shall be packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Respirators and components of respirators, packaged as specified (see 6.2), shall be packed in plywood boxes conforming to PPP-B-601, overseas type. Zinc-coated strapping or strapping of material equivalent in strength and corrosion resistance.

5.2.2 Level B. Respirators and components of respirators, packaged as specified (see 6.2), shall be packed in plywood or fiberboard boxes conforming to PPP-B-601 domestic type, or PPP-B-636, grade V3c.

5.2.3 Level C. Respirators and components of respirators shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, the respirators and its components shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129, as specified (see 6.2).

6. NOTES

6.1 Intended use. (see detailed specification)

6.2 Ordering data. (see detailed specification)

6.2.1 Extra filters and cartridges. Filters and cartridges are not interchangeable between manufacturers nor are they always interchangeable with different models produced by the same manufacturer. Respirators should be purchased with sufficient extra filters and/or cartridges so that the facepiece can be discarded when the filters or cartridges are consumed. In the event that additional filters or cartridges are required, the contracting officer should go back to the same supplier with the exact model, style, and number.

6.3 Chart pattern 20. Chart pattern 20, when resolved under the above condition, indicates as resolving power of 24 seconds. Because the resolving power of the normal eye is approximately 60 seconds, the definition test is not unduly stringent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

6.4 Detail specifications. The following detail specifications form a part of this document:

Detail specification	Title
GGG-M-125/1	Respirator Assemblies: Air Line, with Facepiece (Supplied Air).
GGG-M-125/2	Respirator Assemblies: Air Line, Helmet, (Supplied Air).
GGG-M-125/3	Respirator Assemblies: Air Line, with Disposable Hood, (Supplied Air).
GGG-M-125/4	Respirator Assembly: Half Mask, Air Filtering (Chemical Cartridge).
GGG-M-125/5	Respirator Assembly: Half Mask, Air Filtering (For Metal Fumes, Dusts and Mists).
GGG-M-125/6	Respirator Assembly: Half Mask, Air Filtering; Combination Chemical Cartridge and Filter (For Organic Vapors, Paint Sprays and Dusts).
GGG-M-125/7	Respirator Assembly: Half Mask, Air Filtering (For Pesticides).
GGG-M-125/8	Respirator and Respirator Assembly: Half Mask, Light Weight, Air Filtering (For Pneumoconiosis and Fibrosis Producing Dusts and Nuisance Inert Dusts).
GGG-M-125/9	Respirator Assemblies: Air Line, Hood and Helmet, (For Abrasive Blasting Operations).
GGG-M-125/10	Respirator Assembly: Helmet with Powered-Air-Purifying Unit.
GGG-M-125/11	Respirator Assembly: Helmet with Powered-Air-Purifying Unit (Portable Back Pack).

6.5 The following detailed specification have been issued for respiratory systems. The former classification in GGG-M-125d is indicated:

GGG-M-125/1

Type I, style A	Replaces type I, style A of GGG-M-125d.
Type I, style B	Replaces type I, style B of GGG-M-125d.
GGG-M-125/4	Replaces type II of GGG-M-125d.
GGG-M-125/5	Replaces type III, class 1 of GGG-M-125d.
GGG-M-125/6	Replaces type IV with class 4 filter of GGG-M-125d.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.

GGG-M-125/8	Replaces type III, class 2 of GGG-M-125d.
Type I, style B	
GGG-M-125/9	
Type I, style A	Replaces type V of GGG-M-125d.
Type I, style B	Replaces type V of GGG-M-125d.

6.5.1 GGG-M-125/6 replaces MIL-R-51057A(MU) dated 11 December 1964.

6.6 30 CFR, part 11. 30 CFR, Part 11 supersedes the following Bureau of Mines schedules and approval numbers allowed by the schedules:

19B	Procedure for Testing Supplied-Air Respirators for Permissibility.
21B	Procedure for Testing Filter-Type Dust, Fume and Mist Respirators for Permissibility.
23	Procedure for Testing Non-Emergency Gas Respirators (Chemical Cartridge Respirators) for Permissibility.

MILITARY CUSTODIANS:

Army - MU
Navy - YD
Air Force - 84

Review activities:

Army - MD
Navy - MC

User activities:

Navy - SH, CG

Preparing activity:

Navy - YD

Project No. 4240-0313

Civil Agency Coordinating Activity:

GSA - FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:37Z
Check the source to verify that this is the current version before use.